

Distribution Shift Results - Approx GP

Skin Lesion 26 03 17 16 10 - Full Training Set

Distribution Shift - Scattering (All J, Inv+Cov)

Metric	Base	Scattering	ConvNeXt Atto	ConvNeXt Base	DINOv2 Small	DINOv2 Base
LOSS	0.000 ± 0.000	<u>0.630 ± 0.001</u>	0.741 ± 0.020	0.916 ± 0.035	1.128 ± 0.060	1.054 ± 0.120
MAE	639.6 ± 20.7	<u>350.4 ± 22.9</u>	<u>340.0 ± 18.5</u>	378.1 ± 9.5	533.5 ± 38.4	480.9 ± 32.0
RMSE	817.3 ± 35.0	540.0 ± 33.3	<u>515.4 ± 38.9</u>	555.7 ± 20.5	747.6 ± 56.0	687.4 ± 40.3
NLL	1.501 ± 0.048	1.160 ± 0.117	<u>1.126 ± 0.128</u>	1.148 ± 0.072	1.464 ± 0.119	1.396 ± 0.098
QCE 95	-0.029 ± 0.011	-0.084 ± 0.019	-0.076 ± 0.022	<u>-0.060 ± 0.018</u>	-0.100 ± 0.017	-0.096 ± 0.019
PI 95 μ	3.920 ± 0.000	<u>1.844 ± 0.009</u>	1.926 ± 0.007	2.321 ± 0.051	2.713 ± 0.082	2.500 ± 0.086
PI 95 σ	0.000 ± 0.000	0.112 ± 0.003	0.086 ± 0.010	0.068 ± 0.008	0.129 ± 0.014	<u>0.172 ± 0.039</u>

Histology Nuclei 26 03 17 15 09 - Full Training Set

Distribution Shift - Scattering (All J, Inv+Cov)

Metric	Base	Scattering	ConvNeXt Atto	ConvNeXt Base	DINOv2 Small	DINOv2 Base
LOSS	0.000 ± 0.000	1.040 ± 0.000	0.970 ± 0.029	1.058 ± 0.033	1.036 ± 0.073	0.897 ± 0.007
MAE	540.8 ± 18.0	287.8 ± 12.7	328.7 ± 12.6	335.7 ± 21.9	307.7 ± 12.5	285.5 ± 13.4
RMSE	623.4 ± 14.1	375.1 ± 12.4	450.6 ± 24.8	480.5 ± 25.5	421.1 ± 19.2	392.0 ± 19.5
NLL	1.308 ± 0.018	0.819 ± 0.024	0.867 ± 0.025	0.920 ± 0.028	0.843 ± 0.034	0.774 ± 0.043
QCE 95	0.047 ± 0.003	0.023 ± 0.010	-0.028 ± 0.013	-0.028 ± 0.012	0.002 ± 0.007	0.002 ± 0.014
PI 95 μ	3.920 ± 0.000	2.557 ± 0.003	2.448 ± 0.028	2.580 ± 0.026	2.545 ± 0.031	2.357 ± 0.026
PI 95 σ	0.000 ± 0.000	0.018 ± 0.002	0.052 ± 0.007	0.057 ± 0.002	0.105 ± 0.033	0.121 ± 0.013

Poverty 26 03 18 07 02 - Full Training Set

Distribution Shift - Scattering (All J, Inv+Cov)

Metric	Base	Scattering	ConvNeXt Atto	ConvNeXt Base	DINOv2 Small	DINOv2 Base
LOSS	0.000 $\pm$ 0.000	1.137 $\pm$ 0.010	0.874 $\pm$ 0.001	0.942 $\pm$ 0.035	0.892 $\pm$ 0.002	0.868 $\pm$ 0.002
MAE	0.798 $\pm$ 0.025	0.450 $\pm$ 0.035	0.398 $\pm$ 0.007	0.386 $\pm$ 0.027	0.385 $\pm$ 0.007	0.381 $\pm$ 0.007
RMSE	0.903 $\pm$ 0.020	0.557 $\pm$ 0.037	0.498 $\pm$ 0.007	0.496 $\pm$ 0.027	0.493 $\pm$ 0.010	0.487 $\pm$ 0.013
NLL	1.541 $\pm$ 0.027	0.961 $\pm$ 0.046	0.864 $\pm$ 0.023	0.884 $\pm$ 0.046	0.866 $\pm$ 0.027	0.865 $\pm$ 0.030
QCE 95	0.014 $\pm$ 0.009	0.010 $\pm$ 0.012	<u>-0.007 $\pm$ 0.010</u>	-0.008 $\pm$ 0.014	-0.008 $\pm$ 0.013	-0.010 $\pm$ 0.012
PI 95 μ	3.920 $\pm$ 0.000	2.787 $\pm$ 0.060	2.297 $\pm$ 0.006	2.415 $\pm$ 0.020	2.340 $\pm$ 0.005	2.283 $\pm$ 0.008
PI 95 σ	0.000 $\pm$ 0.000	0.096 $\pm$ 0.026	0.076 $\pm$ 0.002	0.033 $\pm$ 0.003	0.037 $\pm$ 0.001	0.043 $\pm$ 0.002